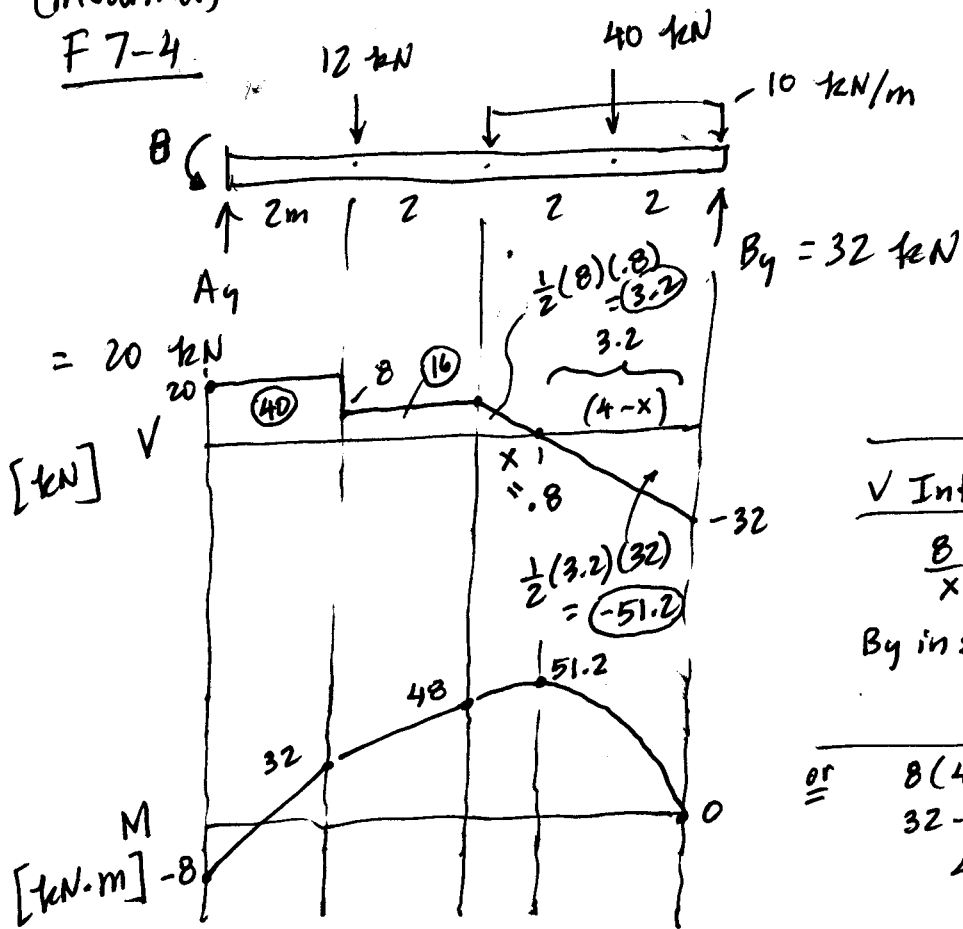


(Modified)

F 7-4



Supports:

$$\begin{aligned} +\uparrow \sum M_A &= 0; \\ -8 + 12(2) + 40(6) - B_y(8) &= 0 \\ B_y &= 32 \text{ kN} \\ A_y &= 20 \text{ kN} \end{aligned}$$

V Intercept:

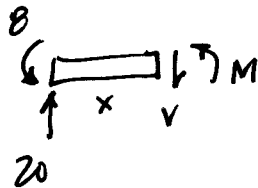
$$\frac{8}{x} = \frac{32}{(4-x)}$$

$$\begin{aligned} \text{By inspection: } x &= .8 \\ 4-x &= 3.2 \end{aligned}$$

$$\begin{aligned} \text{or } 8(4-x) &= 32x \\ 32 - 8x &= 32x \\ 40x &= 32 \\ x &= 32/40 = 4/5 = .8 \end{aligned}$$

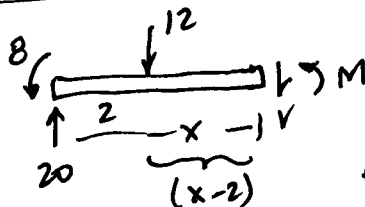
Equations:

$$0 \leq x \leq 2$$



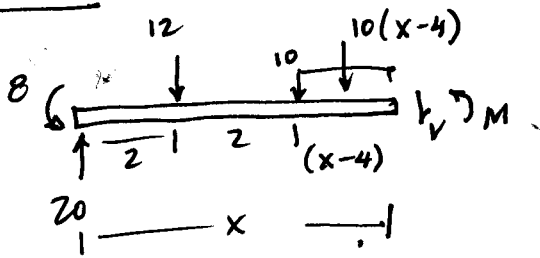
$$\begin{aligned} +\uparrow 20 - V &= 0; \quad V = 20 \text{ kN} \\ +\curvearrowright \text{cut: } M + 8 - 20x &= 0 \\ M &= 20x - 8 \text{ kN}\cdot\text{m} \end{aligned}$$

$$2 \leq x \leq 4$$



$$\begin{aligned} +\uparrow 20 - 12 - V &= 0 \\ V &= 8 \text{ kN} \\ +\curvearrowright \sum M_{\text{cut}} &= 0; \quad M + 8 + 12(x-2) - 20(x) = 0 \\ M &= 20x - 12(x-2) - 8 = 0 \\ M &= 20x - 12x + 24 - 8 = 0 \\ M &= 8x + 16 \text{ kN}\cdot\text{m} \end{aligned}$$

$$4 \leq x \leq 8 \quad (\text{From LHS})$$



$$+\uparrow \quad 20 - 12 - 10(x-4) - V = 0$$

$$V = 8 - 10(x-4)$$

$$V = 8 - 10x + 40$$

$$V = 48 - 10x \text{ kN}$$

$$+\circlearrowleft \sum M_{cut} = 0;$$

$$8 - 20(x) + 12(x-2)$$

$$+ 10(x-4)\left(\frac{x-4}{2}\right) + M = 0$$

$$M = -5(x-4)^2 + 20x - 12(x-2) - 8$$

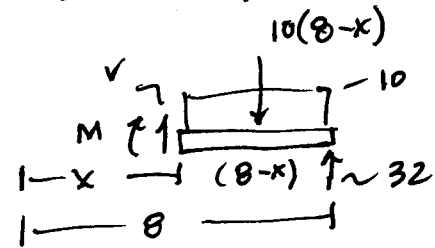
$$\text{at } x=8, \quad M = -5(4)^2 + 160 - 72 - 8$$

$$= -80 + 160 - 72 - 8$$

$$M=0 \text{ at } x=8 \text{ m}$$

Checks!

(RHS FBD)



$$+\uparrow \quad V - 10(8-x) + 32 = 0$$

$$V = 10(8-x) - 32$$

$$V = 80 - 10x - 32$$

$$V = 48 - 10x \text{ kN}$$

$$+\circlearrowleft \sum M_{cut} = 0;$$

$$M + 10(8-x)\left(\frac{8-x}{2}\right)$$

$$- 32(8-x) = 0$$

$$M = 32(8-x) - 5(8-x)^2$$

Check: at $x=8 \text{ m}$

$$M=0 \text{ Checks!}$$

Algebra:

$$M = -5(x^2 - 8x + 16) + 20x$$

$$- 12x + 24 - 8$$

$$M = -5x^2 + 40x - 80 + 8x + 16$$

$$M = -5x^2 + 48x - 64 \text{ kN}\cdot\text{m}$$

Note: $V = -10x + 48$

SAME!

$$M = \int V dx = -5x^2 + 48x + \text{Const}$$

$$M = 256 - 32x$$

$$- 5(64 - 16x + x^2)$$

$$M = 256 - 32x$$

$$- 320 + 80x - 5x^2$$

$$M = -5x^2 + 48x - 64 \text{ kN}\cdot\text{m}$$